HW4

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Question 1

Approach 1

Fully quasi-steady state: (Case IV derived in class)

$r_1 = r_2 \Rightarrow k_1 P_A' \theta_* - k_{-1}\theta_A = k_2\theta_A - k_{-2}\theta_B$

$r_2 = -r_3 \Rightarrow k_2\theta_A - k_{-2}\theta_B = k_{-3}\theta_B - k_3 P_B'\theta_*$

$\theta_A + \theta_B + \theta_C = 1$

QE steps 1 & 3:

$(k_2 \sim k_{-2}) << (k_{-1} \sim k_1) \sim (k_3 \sim k_{-3})$

Dividing the equations by k_1:

$$\frac{k_1 P_A' \theta_* - k_{-1}\theta_A}{k_1} = \frac{k_2}{k_1}\theta_A - \frac{k_{-2}}{k_1}\theta_B \quad (\theta_A \to 0,\ \theta_B \to 0)$$

$$P_A'\theta_* - \frac{k_{-1}}{k_1}\theta_A = 0$$

$$\boxed{\theta_A = \frac{k_1}{k_{-1}} P_A' \theta_*}$$

Similarly,

$$\frac{k_2}{k_1}\theta_A - \frac{k_{-2}}{k_1}\theta_B = \frac{k_{-3}}{k_1}\theta_B - \frac{k_3}{k_1} P_B'\theta_* \quad (\theta_A \to 0,\ \theta_B \to 0)$$

$$0 = k_{-3}\theta_B - k_3 P_B'\theta_*$$

$$0 = \frac{k_{-3}}{k_3}\theta_B - P_B'\theta_*$$

$$\therefore \boxed{\theta_B = \frac{k_3}{k_{-3}} P_B'\theta_*}$$

The obtained θ_A and θ_B expressions are equal to those obtain when step 1 and step 3 are at QE

↓

This much mean that r_1, r_3 are QE

as shown in case 1 in class

Approach 2

(a) dimensional rate expressions:

$r_1 = r_2 \Rightarrow k_1 P_A' \theta_* - k_{-1}\theta_A = k_2\theta_A - k_{-2}\theta_B$

$r_2 = -r_3 \Rightarrow k_2\theta_A - k_{-2}\theta_B = k_{-3}\theta_B - k_3 P_B'\theta_*$

$\theta_A + \theta_B + \theta_C = 1$

Divide by k_{-1}

$r_1 = r_2 \Rightarrow \frac{k_1}{k_{-1}} P_A'\theta_* - \theta_A = \frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B \quad \cdots (1)$

$r_2 = -r_3 \Rightarrow \frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B = \frac{k_{-3}}{k_{-1}}\theta_B - \frac{k_3}{k_{-1}} P_B'\theta_* \quad \cdots (2)$

↑ Non-dimensional equations

(b) $\theta_A = \theta_{A_1} + \theta_{A_2}$

$\theta_B = \theta_{B_1} + \theta_{B_2}$

$\theta_* = \theta_{*_1} + \theta_{*_2}$

Plugging in to the non-dimensional equations:

(1) $\frac{k_1}{k_{-1}} P_A'(\theta_{*_1} + \theta_{*_2}) - (\theta_{A_1} + \theta_{A_2}) = \frac{k_2}{k_{-1}}(\theta_{A_1} + \theta_{A_2}) - \frac{k_{-2}}{k_{-1}}(\theta_{B_1} + \theta_{B_2})$

$\Rightarrow \frac{k_1}{k_{-1}} P_A'\theta_{*_1} - \theta_{A_1} + \frac{k_1}{k_{-1}} P_A'\theta_{*_2} - \theta_{A_2} = \frac{k_2}{k_{-1}}\theta_{A_1} - \frac{k_{-2}}{k_{-1}}\theta_{B_1} + \frac{k_2}{k_{-1}}\theta_{A_2} - \frac{k_{-2}}{k_{-1}}\theta_{B_2}$

$\Rightarrow \boxed{\frac{k_1}{k_{-1}} P_A'\theta_{*_1} + \theta_{A_1}\left(-1 - \frac{k_2}{k_{-1}}\right) + \frac{k_{-2}}{k_{-1}}\theta_{B_1} = \theta_{A_2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}} P_A'\theta_{*_2} - \frac{k_{-2}}{k_{-1}}\theta_{B_2}}$

$\theta_{A_2} << \theta_{A_1}$

$\theta_{B_2} << \theta_{B_1}$

$\theta_{*_2} << \theta_{*_1}$

(2) $\frac{k_2}{k_{-1}}\theta_A - \frac{k_{-2}}{k_{-1}}\theta_B = \frac{k_{-3}}{k_{-1}}\theta_B - \frac{k_3}{k_{-1}} P_B'\theta_*$

$\frac{k_2}{k_{-1}}(\theta_{A_1} + \theta_{A_2}) - \frac{k_{-2}}{k_{-1}}(\theta_{B_1} + \theta_{B_2}) = -\frac{k_{-3}}{k_{-1}}(\theta_{B_1} + \theta_{B_2}) - \frac{k_3}{k_{-1}} P_B'(\theta_{*_1} + \theta_{*_2})$

$$\frac{k_1}{k_{-1}}\theta'_{A1} - \frac{k_{-1}}{k_1}\theta'_{B1} + \frac{k_{-2}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_2}{k_{-1}}\theta'_{A2} + \frac{k_2}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}P_B'$$

③ $\theta_{A1} + \theta_{A2} + \theta'_{B1} + \theta'_{B2} + \theta'_{*1} + \theta'_{*2} = 1$

$$(\theta_{A1} + \theta_{B1} + \theta_{*1}) + (\theta'_{A2} + \theta'_{B2} + \theta'_{*2}) = 1$$

(c) From approach #1: $(k_2 \sim k_{-2}) << (k_1 \sim k_{-1}) \sim (k_3 \sim k_{-3})$

① $\frac{k_1}{k_{-1}}P_A'\theta'_{*1} + \theta_{A1}\left(-1 - \frac{k_2}{k_{-1}}\right) + \frac{k_{-2}}{k_{-1}}\theta'_{B1} = \theta'_{A2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}}P_A'\theta'_{*2} - \frac{k_{-2}}{k_{-1}}\theta'_{B2}$

$$\frac{k_1}{k_{-1}}P_A'\theta'_{*1} - \theta_{A1} = \theta'_{A2} - \frac{k_1}{k_{-1}}P_A'\theta'_{*2}$$

For order 1: $\frac{k_1}{k_{-1}}P_A'\theta'_{*1} - \theta_{A1} = 0 \Rightarrow \theta_{A1} = \frac{k_1}{k_{-1}}P_A'\theta'_{*1}$

② $\frac{k_2}{k_{-1}}\theta'_{A1} - \frac{k_{-1}}{k_1}\theta'_{B1} + \frac{k_{-2}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_2}{k_{-1}}\theta'_{A2} + \frac{k_2}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}$

$$\frac{k_{-2}}{k_{-1}}\theta'_{B1} + \frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2}$$

For order 1: $\frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B1}$

③ For site balance: $\underbrace{(\theta_{A1} + \theta_{B1} + \theta_{*1})}_{=1} + \underbrace{(\theta'_{A2} + \theta'_{B2} + \theta'_{*2})}_{=0} = 1$

$\theta_{A1} + \theta_{B1} + \theta_{*1} = 1$

∴ For first order:

$\theta_{A1} = \frac{k_1}{k_{-1}}P_A'\theta'_{*1}$; $\frac{k_3}{k_{-1}}P_B'\theta'_{*1} = -\frac{k_{-2}}{k_{-1}}\theta'_{B1}$

$$\frac{k_1}{k_{-1}}P_A'\theta'_{*1} + \frac{k_3}{k_{-3}}P_B'\theta'_{*1} + \theta_{*1} = 1$$

$$\theta_{*1} = \frac{1}{\left(\frac{k_1}{k_{-1}}P_A' + \frac{k_3}{k_{-3}}P_B' + 1\right)}$$

This is the first order approximation

For second-order expression, we cannot make the simplification assumptions → Therefore have to solve for the earlier solution.

For second order: 1) $0 = \theta'_{A2}\left(1 + \frac{k_2}{k_{-1}}\right) - \frac{k_1}{k_{-1}}P_A'\theta'_{*2} - \frac{k_{-2}}{k_{-1}}\theta'_{B2}P_B' \Rightarrow \theta'_{A2}(k_{-1} + k_2) + k_1P_A'\theta'_{*2} - k_{-2}\theta'_{B2}P_B' = 0 \Rightarrow \theta'_{B2} = \frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{P_B'k_{-2}}$

2) $0 = -\frac{k_2}{k_1}\theta'_{A2} + \frac{k_{-2}}{k_{-1}}\theta'_{B2} - \frac{k_{-3}}{k_{-1}}\theta'_{B2} - \frac{k_3}{k_{-1}}\theta'_{*2} \Rightarrow -k_2\theta'_{A2} + k_{-2}\theta'_{B2} - k_{-3}\theta'_{B2} - k_3\theta'_{*2} = 0 \Rightarrow -k_2\theta'_{A2} + (k_{-2} - k_{-3})\left(\frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{P_B'k_{-2}}\right) - k_3\theta'_{*2} = 0$

$$\Rightarrow -k_2\theta'_{A2} + \underbrace{\frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B'k_{-2}}}_{A_1}\theta'_{A2} + \underbrace{\frac{(k_{-2}-k_{-3})}{k_{-2}}}_{A_2}k_1P_A'\theta'_{*2} - k_3\theta'_{*2} = 0$$

$$\theta'_{A2}\left(-k_2 + \frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B'k_{-2}}\right) + \theta'_{*2}\left[\frac{(k_{-2}-k_{-3})}{k_{-2}}k_1P_A' - k_3\right] = 0$$

$$\theta'_{A2} = \frac{-\theta'_{*2}[A_2]}{(A_1)}$$

3) $\theta'_{A2} + \theta'_{B2} + \theta'_{*2} = 0$

Plugging everything back in expression ②:

$$-\frac{\theta'_{*2}[A_2]}{[A_1]} + \frac{\theta'_{A2}(k_{-1}+k_2) + k_1P_A'\theta'_{*2}}{k_{-2}} + \theta'_{*2} = 0$$

$$-\frac{\theta_{*2}[A_2]}{[A_1]} - \frac{(k_{-1}+k_2)}{k_{-2}}\frac{[A_2]}{[A_1]}\theta_{*2} + \frac{k_1 P_A'\theta_{*2}}{k_{-2}} + \theta_{*2} = 0 \quad \Rightarrow \boxed{\theta_{*2} = 0}$$

Plugging in θ_{*2} will give $\theta_{B2} = f(\theta_{A2})$ only!

$$\therefore \quad \theta_{A_2} = -\theta_{*2}\frac{\left[\frac{(k_{-2}-k_{-3})}{k_{-2}}\frac{k_1 P_A'}{} - k_3\right]}{\left[-k_2 + \frac{(k_{-2}-k_{-3})(k_{-1}+k_2)}{P_B' k_{-2}}\right]} \quad ; \quad \theta_{B2} = \frac{\theta_{A_2}(k_{-1}+k_2) + k_1 P_A'\theta_{*2}}{P_B' k_{-2}}$$

(d) $r_1 = r_2 = -r_3$

$r_2 = k_2\theta_A - k_{-2}\theta_B$

For first order:

$$r_2^{(1)} = k_2\theta_{A1} - k_{-2}\theta_{B1} \quad ; \quad \theta_{A1} = \frac{k_1}{k_{-1}}P_A'\theta_{*1} \quad ; \quad \frac{k_3}{k_{-1}}P_B'\theta_{*1} = -\frac{k_{-3}}{k_{-1}}\theta_{B1}$$

$$= \frac{k_2 k_1}{k_{-1}}P_A'\theta_{*1} - k_{-2}\frac{k_3}{k_{-3}}P_B'\theta_{*1}$$

$$r_2^{(1)} = \left[k_1 K_1 P_A' - k_{-2}K_3 P_B'\right]\theta_{*1} \quad ; \quad \theta_{*1} = \frac{1}{\left(\frac{k_1}{k_{-1}}P_A' + \frac{k_3}{k_{-3}}P_B' + 1\right)}$$

This first order approximation matches the QE expressions obtained in approach 1 → Exact solution

(e) First order solution represents step 1 and step 3 as QE

↳ This means that the forward & reverse rates are much faster than that of step 2 so it therefore "appears" that step 1 & 2 are at equilibrium

↳ The first order perturbation represents that exact qE solution

↳ Given this analysis, the net rate should be first order (since second order solution gives the trivial case)

